
Enhanced Multi-Instance Partial Label Learning via Average Gradient Outer Product

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Abstract

Multi-Instance Partial-Label Learning (MIPL) is a *dual* weakly supervised framework where each training bag is annotated with a candidate label set comprising a single ground-truth label and multiple false positives. The core challenge in MIPL is that the association between key instances and the ground-truth label is significantly obscured by these coarse and ambiguous supervision. Existing methods typically rely on model predictions to aggregate instance features via learned attention weights; however, this approach is susceptible to false positives, often causing key instances to be overlooked. To address this, we propose Average Gradient Outer Product based Multi-Instance Partial-Label Learning (AGOP-MIPL). Specifically, we compute the Average Gradient Outer Product (AGOP) from the gradients of the bag-level classifier, integrating it into an attention module to amplify discriminative feature directions and facilitate key-instance identification. Furthermore, we introduce feature prototypes and a progressive disambiguation strategy to further suppress noisy candidates. Extensive experiments on four MIPL benchmarks and a real-world CRC-MIPL dataset demonstrate that AGOPMIPL consistently outperforms five state-of-the-art baselines, achieving a relative gain of up to 25.9% on CRC-MIPL-KMeansSeg.

1. Introduction

While supervised learning has achieved remarkable success, its reliance on exhaustive, high-quality annotations presents a severe bottleneck due to prohibitive labeling costs. This

limitation has catalyzed the rise of weakly supervised learning (Zhou, 2018; Li et al., 2019; Nodet et al., 2021). Among these paradigms, Multi-Instance Learning (MIL) (Dietterich et al., 1997; Zhou et al., 2009; Zhang & Jin, 2020) has emerged as a prominent framework utilizing coarse bag-level labels. However, standard MIL often proves insufficient for complex real-world scenarios where supervision is not only coarse but also ambiguous. Consider image annotation via crowdsourcing: an image (bag) containing multiple objects often yields a candidate label set comprising both the ground truth and false positives. This scenario introduces a dual weak supervision: uncertainty regarding which instances are informative (instance space) and which label within the candidate set is correct (label space). To address this compounded challenge, Multi-Instance Partial-Label Learning (MIPL) (Tang et al., 2023) has been proposed, aiming to jointly disambiguate the label space and localize key instances under such weak supervision.

Existing MIPL methods generally fall into two categories, and both largely inherit their machinery from MIL and partial-label learning rather than addressing how the two ambiguities interact. Instance-level methods propagate the bag’s candidate set down to each instance and disambiguate locally (Tang et al., 2024b), which relocates the partial-label problem onto the instances but forfeits the global bag context. Bag-level methods instead aggregate instances into a bag representation and disambiguate at the bag level (Tang et al., 2023; 2024a; Yang et al., 2024; 2025); crucially, the aggregation weights are read off inter-instance correlations or distributional priors, so the candidate labels supervise only the bag-level classifier and never tell the attention which feature directions distinguish the true label. Being label-agnostic, the attention simply follows the most salient or mutually correlated instances. This is precisely why noisy candidates are damaging: the false-positive labels corrupt the bag-level supervision, while the label-blind attention has no way to recover the true key instance, so attention drifts onto irrelevant ones and the key instance carrying the true label is overlooked. To validate this limitation, we evaluated four instance-attention-based MIPL methods on the FMNIST-MIPL dataset under varying noise ratios r . As shown in Figure 1, existing methods demonstrate limited efficacy in identifying key instances. Specifically,

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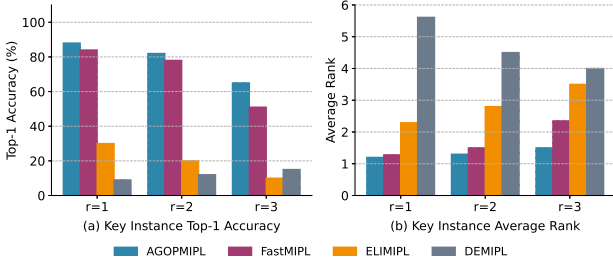


Figure 1. Key instance identification on FMNIST-MIPL with varying noise ratios r . (a) Top-1 hit rate. (b) Average rank under descending attention scores (lower is better).

Figure 1(a) reveals that the accuracy of selecting the correct key instance declines as label noise increases. Similarly, Figure 1(b), which depicts the average attention ranking of key instances (where a lower rank indicates higher attention), confirms that state-of-the-art methods possess a limited capacity to distinguish truly informative instances.

To address these challenges, we propose AGOPMIPL (Average Gradient Outer Product based Multi-Instance Partial-Label learning). Our approach is motivated by a pivotal observation: the gradients of the classifier with respect to the input contain rich information regarding discriminative feature directions, which remain robust even under label noise. Specifically, we utilize the average outer product of the classifier’s Jacobian to explicitly identify these directions. By employing an AGOP-induced metric transformation, we rescale the feature space to amplify dimensions critical for classification. This transformation effectively suppresses noisy interference and steers the attention mechanism toward truly informative instances. Complementing this, we incorporate a prototype-based attention mechanism and a progressive label disambiguation module, ensuring the robust identification of both key instances and ground-truth labels. Our contributions are fourfold:

- We propose an AGOP-based feature learning method that dynamically isolates discriminative features, enhancing the model’s resilience to label ambiguity.
- We develop an attention mechanism that integrates AGOP-derived feature metrics with class prototypes, ensuring attention scores reflect both instance saliency and semantic alignment.
- We formulate a progressive disambiguation strategy that iteratively refines the label confidence distribution, enabling precise candidate label selection during training.
- We provide a theoretical analysis demonstrating that the AGOP transformation recovers the underlying discriminative subspace, theoretically guaranteeing the amplification of features aligned with the true label.

The remainder of this paper is organized as follows. Section 2 reviews related works. Section 3 details the proposed method. Section 4 presents a theoretical analysis of AGOP’s effectiveness in MIPL. Section 5 reports empirical results, and Section 6 concludes the paper.

2. Related Works

2.1. Multi-Instance Learning

Multi-instance learning (MIL) (Dietterich et al., 1997) deals with the problem of learning from bags of instances using only bag-level supervision. Existing literature is broadly categorized into two streams: instance-level methods and bag-level methods (Amores, 2013; Carbonneau et al., 2018).

Instance-level methods aim to learn a classifier capable of assigning labels to individual instances. This classifier is subsequently integrated with an MI assumption that links instance labels to bag labels. Numerous strategies originally developed for single-instance learning have been adapted to construct such instance classifiers. For instance, Diverse Density (DD) (Maron & Lozano-Pérez, 1998) and its variant EM-DD (Zhang & Goldman, 2001) employ maximum likelihood estimation. ID3-MI (Chevaleyre & Zucker, 2001) adapts the standard ID3 decision tree algorithm, while MI-SVM (Andrews et al., 2002) and MI-ODM (Zhang & Jin, 2020) are derived by optimizing the minimum margin and margin distribution, respectively. Furthermore, BP-MIP (Zhou & Zhang, 2002), EM-CNN (Hou et al., 2016), and SNL (Garcez & Zaverucha, 2012) leverage Feedforward Neural Networks (FNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) architectures.

Bag-level methods predict bag labels by treating each training bag as a holistic entity, thereby bypassing the inference of latent instance labels. The first approach defines bag-wise distance or similarity functions, enabling the application of nearest neighbor and kernel methods; Citation-KNN (Wang & Zucker, 2000) and the MI Kernels (Gärtner et al., 2002) are representative examples. The second approach maps each bag to a single feature representation, allowing the use of single-instance learning algorithms. This mapping is typically achieved via statistical aggregation (e.g., mean, min, max, and moments) (Amores, 2013) or instance feature aggregation (Wei et al., 2017).

2.2. Multi-Instance Partial-Label Learning

Multi-instance partial-label learning (MIPL) (Tang et al., 2023) inherits the instance-level and bag-level dichotomy of MIL; the twist contributed by the partial-label side (Cour et al., 2011) is that the bag is now supervised by a candidate set rather than a single label, so each stream must disambiguate the candidate labels on top of locating key instances.

Notably, most methods meet this twist by recombining established components rather than by confronting how the two ambiguities interact.

Instance-level methods propagate the bag’s candidate set down to every instance and resolve it there. MIPLGP (Tang et al., 2024b) assigns the candidate set to each instance, disambiguates it into a soft per-instance label, and pools the resulting instance predictions into the bag label under the multi-instance assumption. This neatly recasts the bag as a collection of partial-label examples, but reasoning over isolated instances forfeits the global bag context that is often what makes an ambiguous instance interpretable in the first place.

Bag-level methods compress the bag into a single representation and only then confront the candidate set, which raises the question of whether the candidate labels ever shape that representation. They rarely do. DEMIPL (Tang et al., 2023) and ELIMIPL (Tang et al., 2024a) pool instances with an attention whose weights are read off inter-instance correlations—exactly as in ordinary MIL—and reintroduce the candidate set only at the output, through a momentum-based or a sparsity-and-inhibition disambiguation inherited from partial-label learning; the bag-level model thus largely amounts to MIL aggregation followed by partial-label disambiguation. FastMIPL (Yang et al., 2025) and ProMIPL (Yang et al., 2024) wrap the same aggregation in a probabilistic model—importance pooling under mixed-effects priors and a generative model of bags and candidate labels, respectively—yet the candidate supervision still reaches the aggregation only indirectly, through the likelihood or the variational objective. In every case, the weights that decide which instances constitute the bag are set by instance statistics, correlations, or distributional priors, never by the feature directions that are discriminative for the true label. The noisy candidate set therefore seeps unchecked into key-instance localization, which is exactly where these methods prove most fragile.

This is the gap AGOPMIPL targets: rather than letting the candidate labels reach the aggregation only after the fact, it reads the label-discriminative directions directly off the classifier through the average gradient outer product and reshapes the feature metric before attention, keeping key-instance localization aligned with the true label and robust to candidate-set noise.

2.3. Average Gradient Outer Product

The Average Gradient Outer Product (AGOP) (Radhakrishnan et al., 2024) has recently emerged as a principled tool for feature learning. For a predictor $f : \mathbb{R}^d \mapsto \mathbb{R}^c$, the Jacobian matrix $\mathbf{J}_f(\mathbf{x}) \in \mathbb{R}^{c \times d}$ captures the local sensitivity of the output with respect to input perturbations. The AGOP is

defined as the average outer product of the Jacobian:

$$\mathbf{G} = \frac{1}{|\mathcal{X}|} \sum_{\mathbf{x} \in \mathcal{X}} \mathbf{J}_f(\mathbf{x})^\top \mathbf{J}_f(\mathbf{x}) \in \mathbb{R}^{d \times d}.$$

Intuitively, the diagonal entries of $\mathbf{G}$ quantify the average squared sensitivity of the outputs to individual features, thereby identifying the most discriminative dimensions. Conversely, the off-diagonal entries reveal feature co-importance by capturing correlations between feature directions.

Beyond identifying discriminative features, AGOP has been established as a unifying mechanism characterizing feature learning across diverse architectures, including MLP, CNN, RNN and Transformer. Building on this foundation, the Recursive Feature Machine (RFM) (Radhakrishnan et al., 2024) leverages AGOP for explicit feature learning. Specifically, RFM transforms the input space via $\mathbf{x} \mapsto \mathbf{G}^{1/2} \mathbf{x}$. This mapping effectively rescales the feature space along informative directions, implementing a learned Mahalanobis metric. Theoretically, this transformation is grounded in Riemannian geometry: the AGOP defines a metric tensor, and $\mathbf{G}^{1/2}$ maps the original Euclidean space to a geometry where Euclidean distances align with the learned metric (Radhakrishnan et al., 2025). Despite its demonstrated efficacy in fully supervised settings, the potential of AGOP in weakly supervised scenarios remains largely underexplored.

3. Proposed Method

3.1. Preliminaries

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the input instance space and $\mathcal{Y} = [q] \triangleq \{1, 2, \dots, q\}$ be the label set. We consider a MIPL dataset $\mathcal{D} = \{(\mathbf{X}_i, \mathcal{S}_i)\}_{i \in [m]}$ comprising m bags, where each bag $\mathbf{X}_i = \{\mathbf{x}_i^{(j)}\}_{j \in [n_i]}$ consists of n_i instances. The ground-truth label Y_i is unknown but guaranteed to belong to the candidate label set $\mathcal{S}_i \subseteq \mathcal{Y}$; we write $s_{i,c} = \mathbb{I}(c \in \mathcal{S}_i)$ for the candidate indicator, where $\mathbb{I}(\cdot)$ is the indicator function.

For model training, we associate each bag $\mathbf{X}_i$ with a soft label vector $\mathbf{y}_i \in \Delta_{q-1}$ where Δ_{q-1} is the $(q-1)$ dimensional simplex, and $\mathbf{y}_i$ is typically initialized by uniformly distributing probability mass across the candidate set:

$$y_{i,c} = s_{i,c}/|\mathcal{S}_i|, \quad c \in [q], \quad (1)$$

An overview of the AGOPMIPL framework is illustrated in Figure 2. The framework is primarily composed of three components. First, the instance prototype construction module employs the k -means clustering algorithm to derive cluster centroids, which serve as instance prototypes for each class label. Second, the AGOP feature learning module computes the Average Gradient Outer Product (AGOP) matrix, denoted as $\mathbf{G}_r$, in the r -th round based on the current

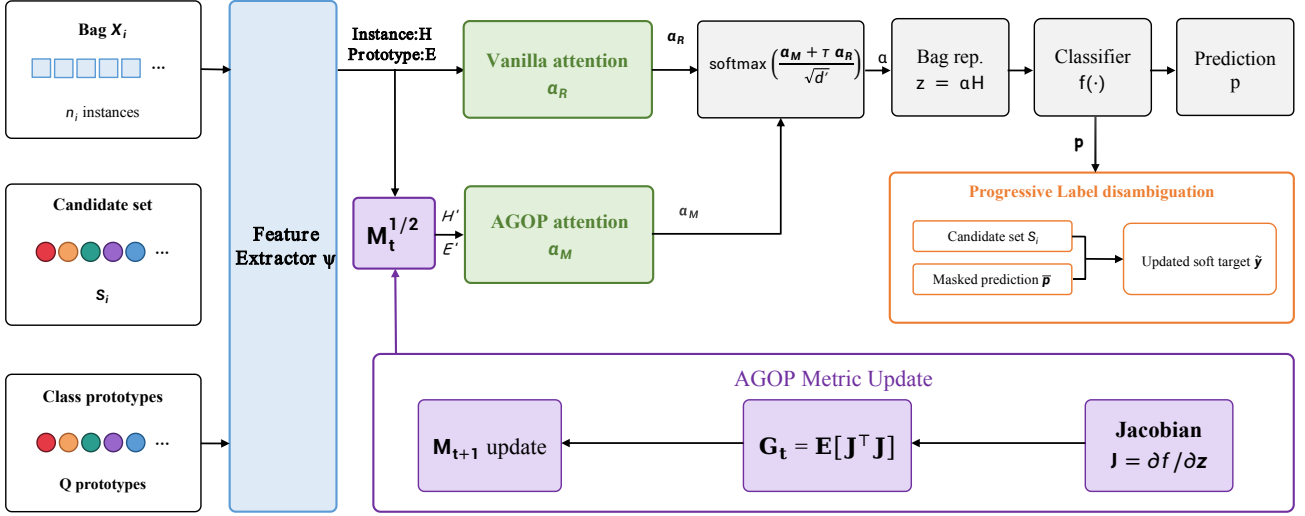


Figure 2. Overview of the AGOPMIPL framework. M_{r-1} is the feature-importance matrix carried into the r -th AGOP round; $H' = HM_{r-1}^{1/2}$ and $E' = EM_{r-1}^{1/2}$ are the AGOP-transformed instance and prototype features, and after the round the AGOP G_r updates it to M_r ; α_R and α_M are the vanilla attention and AGOP attention scores; p is the predicted class probability used in the progressive label disambiguation step to produce the updated soft target $\hat{y}$.

features and classifier. By identifying the directions in the feature space that are most discriminative for classification, G_r is utilized to transform both the instance features H and the prototype features E . Finally, the bag-level feature generation module computes attention scores by leveraging both the original and AGOP-transformed features, subsequently aggregating them to synthesize the final bag-level representations.

3.2. Instance Prototype Generation

Given the MIPL dataset $\mathcal{D} = \{(X_i, S_i)\}_{i \in [m]}$, we construct a class-specific instance pool $\mathcal{P}_c$ for each label c , comprising instances drawn from relevant bags. We then apply the k -means to partition $\mathcal{P}_c$ into n_p clusters, where n_p is the number of prototypes per class (set to q by default). Unlike KISAR (Li et al., 2012), which clusters all instances globally and thus overlooks specific instance-label dependencies, our approach focuses on label-aware prototype generation. The resulting n_p centroids serve as the instance prototypes $P_c = \{c_p\}_{p \in [n_p]}$ for the c -th class, which ensures that the centroids effectively capture the underlying distribution of key instances responsible for triggering the c -th class label.

3.3. Bag-Level Feature Generator

With the instance prototypes established, we proceed to extract feature representations for both instances and prototypes to facilitate bag-level aggregation. For a given bag $X_i = \{x_i^{(j)}\}_{j \in [n_i]}$, a feature extractor $\psi(\cdot)$ produces the instance embeddings $H_i = \psi(X_i) = \{h_i^{(j)}\}_{j \in [n_i]}$, where $h_i^{(j)} \in \mathbb{R}^{d'}$ is the embedding of the j -th instance in the i -th

bag. For the prototypes, the n_p centroids of each class c (i.e., P_c) are embedded and combined into a single representative prototype through a learnable weighting $a \in \mathbb{R}^{n_p}$ shared across classes, $e_c = \sum_{p \in [n_p]} a_p \psi(c_p) \in \mathbb{R}^{d'}$; stacking the q class prototypes yields the prototype matrix $E = [e_1; \dots; e_q] \in \mathbb{R}^{q \times d'}$, whose c -th row is the prototype feature of class c . The architecture of $\psi(\cdot)$ is tailored to the specific data modality, typically consisting of convolutional layers for spatial feature extraction and fully connected layers to project the outputs into a d' -dimensional embedding space.

3.4. Average Gradient Outer Product Update

Noisy candidate labels often misguide attention mechanisms dependent solely on instance correlations. To address this, we employ the AGOP (Radhakrishnan et al., 2024) to identify and emphasize discriminative features. Our key insight is that the classifier’s sensitivity, captured by the Jacobian matrix, highlights features critical for identifying key instances.

Let $f: \mathbb{R}^{d'} \mapsto \mathbb{R}^q$ be the classifier mapping bag features $z \in \mathbb{R}^{d'}$ to class logits. The Jacobian $J(z) \in \mathbb{R}^{q \times d'}$, with the (c, l) -th entry as $\partial f_c(z) / \partial z_l$ capturing the sensitivity of each output to each input feature, is obtained *analytically* from the classifier head, rather than via numerical differentiation or per-instance backpropagation. In our implementation, AGOP uses the logit-level Jacobian of a linear classifier head, so $\partial f_c(z) / \partial z_l$ is exactly the corresponding classifier weight. If one instead uses softmax probabilities, the Jacobian is still available in closed form. Thus, the AGOP update avoids both finite-difference passes and the

storage of per-instance Jacobians. To capture global feature importance across the dataset, we compute the AGOP:

$$\mathbf{G}_r = \frac{1}{m} \sum_{i \in [m]} \mathbf{J}(\mathbf{z}_i)^\top \mathbf{J}(\mathbf{z}_i) \in \mathbb{R}^{d' \times d'} \quad (2)$$

The resulting matrix $\mathbf{G}_r$ has a clear interpretation: the diagonal entry $\mathbf{G}_{ll}$ represents the cumulative importance of the l -th feature dimension across all classes, while the off-diagonal entry $\mathbf{G}_{ll'}$ captures the co-importance between features l and l' in influencing predictions. A further advantage of AGOP in the MIPL setting is that the resulting feature-importance matrix is comparatively robust to label noise. At the population level, true-label-related directions are expected to induce more coherent classifier sensitivities across training bags, whereas directions tied to random false-positive labels are less stable and are attenuated in the second-order sensitivity matrix. This should be understood as attenuation in expectation, rather than exact cancellation sample by sample. A formal version of this argument, together with empirical evidence on the spectrum of $\mathbf{G}_r$, is given in Section 4.

Across rounds, the feature-importance matrix is refreshed by an exponential moving average of the AGOP, and round r reshapes the features with the matrix accumulated through the previous round:

$$\begin{aligned} \mathbf{M}_r &= \beta \mathbf{M}_{r-1} + (1 - \beta) \mathbf{G}_r, \quad \mathbf{M}_0 = \mathbf{I}, \\ \mathbf{H}'_i &= \mathbf{H}_i \mathbf{M}_{r-1}^{1/2}, \quad \mathbf{E}' = \mathbf{E} \mathbf{M}_{r-1}^{1/2}, \end{aligned}$$

where $\beta \in [0, 1]$ is the EMA coefficient. This transformation rescales the feature space: dimensions with high importance (large diagonal entries in $\mathbf{M}$) are amplified, while less informative dimensions are suppressed. The square root transformation ensures that Euclidean distances in the transformed space correspond to Mahalanobis distances under the learned metric $\mathbf{M}$. The matrix square root is computed via eigenvalue decomposition: $\mathbf{M}^{1/2} = \mathbf{V} \text{diag}(\sqrt{\boldsymbol{\lambda}}) \mathbf{V}^\top$, where $\mathbf{V}$ and $\boldsymbol{\lambda}$ are the eigenvectors and eigenvalues of $\mathbf{M}$. We retain both the transformed features ($\mathbf{H}'_i, \mathbf{E}'$) and original features ($\mathbf{H}_i, \mathbf{E}$) to capture complementary information.

3.5. Attention Aggregation

To aggregate instance-level features into a bag-level representation, a dual-branch attention mechanism is utilized that computes attention scores from both the raw features ($\mathbf{H} \in \mathbb{R}^{n_i \times d'}$, $\mathbf{E} \in \mathbb{R}^{q \times d'}$) and the AGOP-transformed features ($\mathbf{H}', \mathbf{E}'$). For brevity, we define the inner attention map

$$g(\mathbf{A}, \mathbf{B}; \mathbf{W}_{\{V, U, 1\}}) = (\sigma(\mathbf{A} \mathbf{W}_V^\top) \odot \sigma(\mathbf{B} \mathbf{W}_U^\top))^\top \mathbf{W}_1^\top,$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\odot$ the Hadamard (element-wise) product broadcast over the n_i instances, and

D the hidden dimension of the attention network. For the vanilla branch, $\mathbf{W}_V \in \mathbb{R}^{q \times d'}$, $\mathbf{W}_U \in \mathbb{R}^{1 \times d'}$, $\mathbf{W}_1 \in \mathbb{R}^{D \times q}$, and $\mathbf{W}_2 \in \mathbb{R}^{1 \times D}$; the AGOP branch uses the primed parameters $\mathbf{W}'_{\{V, U, 1, 2\}}$ with matching shapes. The vanilla attention and AGOP attention scores are then

$$\alpha_R = \mathbf{W}_2 \cdot \text{ReLU}(g(\mathbf{H}, \mathbf{E}; \mathbf{W}_{\{V, U, 1\}}))^\top, \quad (3)$$

$$\alpha_M = \mathbf{W}'_2 \cdot \text{ReLU}(g(\mathbf{H}', \mathbf{E}'; \mathbf{W}'_{\{V, U, 1\}}))^\top, \quad (4)$$

where $\mathbf{H}', \mathbf{E}'$ are the AGOP-transformed instance and prototype features. The score α_M captures importance from AGOP-transformed features, while α_R captures it from raw features.

The final attention weights combine both branches:

$$\alpha = \text{softmax}((\alpha_M + \tau \alpha_R) / \sqrt{d'}) \in \mathbb{R}^{1 \times n_i}, \quad (5)$$

where τ is a scalar that balances the two attention branches and is set to 1.0 in the experiments reported in this paper.

The final bag representation $\mathbf{z}$ is obtained from the *raw* instance features (rather than the AGOP-transformed ones) so that the AGOP step acts only as attention guidance, leaving the aggregated content unaltered:

$$\mathbf{z} = \alpha \mathbf{H}. \quad (6)$$

A bag-representation ablation that compares this default with the AGOP-transformed and concatenated alternatives is reported in Appendix D.

3.6. Progressive Label Disambiguation

In MIPL tasks, the ground-truth label is concealed within the noisy candidate set, and false positive labels can mislead the training process. We propose a progressive disambiguation strategy inspired by curriculum learning (Bengio et al., 2009), where the model initially relies on the candidate labels and gradually shifts trust to its own predictions as training progresses. At the beginning of training, the model’s predictions are unreliable, making it necessary to rely on the provided candidate labels despite their noise. As training progresses and the model learns discriminative patterns, its predictions become more accurate and can help identify false positives. This progressive trust transfer enables the model to bootstrap from noisy supervision toward accurate label disambiguation.

Let $\mathbf{y}_i \in \Delta_{q-1}$ be the candidate-label vector defined in Eqn. (1), and let $\mathbf{p}_i^{(t)} = \text{softmax}(f(\mathbf{z}_i)) \in \Delta_{q-1}$ denote the model’s predicted probability at epoch t , with $p_{i,c}^{(t)}$ its c -th entry. We mask the prediction to the candidate set and form the training target $\tilde{\mathbf{y}}_i^{(t)}$ as a convex combination:

$$\bar{p}_{i,c}^{(t)} = \frac{(p_{i,c}^{(t)} + \epsilon) \cdot s_{i,c}}{\sum_{k \in \mathcal{S}_i} (p_{i,k}^{(t)} + \epsilon)}, \quad \tilde{\mathbf{y}}_i^{(t)} = \rho_t \mathbf{y}_i + (1 - \rho_t) \bar{\mathbf{p}}_i^{(t)},$$

where $\epsilon > 0$ is a small smoothing constant. This normalization keeps all probability mass inside the candidate set. Within each AGOP round of N epochs, the confidence factor ρ_t follows a warmup-then-decay schedule over the epoch index t : it stays at $\rho_t = 1$ for the first $\lfloor \nu N \rfloor$ warmup epochs (we use $\nu = 0.2$) to prevent premature disambiguation, and then decays linearly as $\rho_t = (N - t) / ((1 - \nu)N)$ to reach 0 at the end of the round. The schedule is renewed each round as the AGOP-transformed features are refreshed.

3.7. Optimization Objective

The total objective function integrates three distinct components. It comprises a *candidate mapping loss* $\mathcal{L}_{map} = -\sum_{i \in [m]} \sum_{c \in [q]} \tilde{y}_{i,c} \log(p_{i,c})$, which aligns predictions with the candidate sets via smoothed targets; a *sparsity loss* $\mathcal{L}_{spar} = -\sum_{i \in [m]} \sum_{c \in [q]} \tilde{p}_{i,c} \log(\tilde{p}_{i,c})$, designed to pinpoint the unique ground truth by minimizing the entropy of the candidate-renormalized prediction $\tilde{p}_{i,c}$ defined in Section 3.6; and a *non-candidate inhibition loss* $\mathcal{L}_{in} = -\sum_{i \in [m]} \sum_{c: s_{i,c}=0} \log(1 - p_{i,c})$, which penalizes predictions for labels excluded from the candidate set. The final composite loss is expressed as:

$$\mathcal{L} = \mathcal{L}_{map} + \mu \mathcal{L}_{spar} + \gamma \mathcal{L}_{in}. \quad (7)$$

where $\mu, \gamma > 0$ are trade-off weights that balance the three loss terms.

4. Theoretical Insight

This section explains, from a population perspective, why AGOP is well suited to MIPL. The underlying mechanism is that AGOP does not denoise the candidate labels directly; it reshapes the feature geometry along directions to which the bag-level classifier is consistently sensitive across training bags, and thereby decouples the true-label signal from candidate-set noise *before* attention aggregation.

4.1. Signal-plus-Noise Decomposition of the Jacobian

Let $\mathbf{z} \in \mathbb{R}^{d'}$ denote the bag-level representation and $\mathbf{J}(\mathbf{z}) = \partial f(\mathbf{z}) / \partial \mathbf{z}$ the Jacobian of the bag-level classifier head. For a bag with candidate-set noise, we decompose

$$\mathbf{J}(\mathbf{z}) = \mathbf{J}_{\text{true}}(\mathbf{z}) + \mathbf{J}_{\text{noise}}(\mathbf{z}), \quad (8)$$

where $\mathbf{J}_{\text{true}}$ captures the sensitivity aligned with the key-instance / true-label signal, while $\mathbf{J}_{\text{noise}}$ captures the sensitivity induced by false-positive candidates and irrelevant instances.

Assumption 4.1. The task-relevant and noise-induced Jacobians have the following population second-order structure:

$$\mathbb{E}[\mathbf{J}_{\text{true}}^\top \mathbf{J}_{\text{true}}] = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top, \quad \mathbb{E}[\mathbf{J}_{\text{noise}}^\top \mathbf{J}_{\text{noise}}] = \sigma^2 \mathbf{I},$$

where $\mathbf{U} \in \mathbb{R}^{d' \times s}$ has orthonormal columns spanning a rank- s discriminative subspace, $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_s)$ with $\lambda_1 \geq \dots \geq \lambda_s > 0$, and $\sigma^2 > 0$. We also assume that the two components have zero cross-covariance in expectation.

The first condition says that the stable, task-relevant sensitivity lies mainly in the shared subspace $\text{span}(\mathbf{U})$. The second treats the sensitivity caused by false-positive labels and irrelevant instances as isotropic at the population level. The zero cross-covariance condition rules out systematic alignment between these two components. It does not require every noisy direction in a finite sample to cancel its task-relevant counterpart. The assumption is idealized, but it captures the regime in which candidate-induced variation is mostly unstructured and weakly aligned with $\text{span}(\mathbf{U})$.

4.2. Discriminative-Subspace Recovery and Direction Amplification

Proposition 4.2 (AGOP recovers the discriminative subspace). *Under Assumption 4.1, suppressing the round index, the AGOP matrix admits the decomposition*

$$\mathbf{G} = \mathbb{E}[\mathbf{J}^\top \mathbf{J}] = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top + \sigma^2 \mathbf{I}. \quad (9)$$

Consequently, the top- s eigenspace of $\mathbf{G}$ coincides with the discriminative subspace $\text{span}(\mathbf{U})$, with eigenvalues $\lambda_i + \sigma^2$ ($i = 1, \dots, s$) and a flat noise floor σ^2 on the orthogonal complement. Moreover, for any $\mathbf{x} = \mathbf{x}_U + \mathbf{x}_V$ with $\mathbf{x}_U \in \text{span}(\mathbf{U})$, $\mathbf{x}_V \in \text{span}(\mathbf{U})^\perp$ and $\mathbf{x}_V \neq \mathbf{0}$, the AGOP transform $\mathbf{G}^{1/2}$ satisfies

$$\frac{\|\mathbf{G}^{1/2} \mathbf{x}_U\|}{\|\mathbf{G}^{1/2} \mathbf{x}_V\|} \geq \sqrt{1 + \frac{\lambda_s}{\sigma^2}} \cdot \frac{\|\mathbf{x}_U\|}{\|\mathbf{x}_V\|}. \quad (10)$$

The proof is deferred to Appendix B. Equation (10) explains why AGOP helps in noisy MIPL: the transform $\mathbf{G}^{1/2}$ stretches true-label-aligned directions relative to false-positive- and noise-aligned directions by a factor of at least $\sqrt{1 + \lambda_s / \sigma^2}$. This amplification occurs *before* attention aggregation, so attention is computed in a geometry where candidate-noise directions are less dominant. The amplification factor grows with λ_s / σ^2 , i.e., with the relative strength of the discriminative signal compared with candidate-set noise.

5. Experiments

To demonstrate the superiority of the proposed AGOPMIPL method, comparative experiments are conducted with other MIPL methods on multiple datasets, including benchmark datasets and real-world datasets.* Furthermore, ablation studies are performed to verify the effectiveness of the

*Our implementation is available at <https://github.com/LittleSpecial/AGOPMIPL>.

AGOP module, and sensitivity analysis of the hyperparameters is also carried out.

Four commonly used benchmark MIPL datasets are utilized, i.e., MNIST-MIPL, FMNIST-MIPL, BIRDSONG-MIPL, and SIVAL-MIPL from the domains of image and biology. Table 1 summarizes the characteristics of both the benchmark and real-world datasets. #bags, #ins, #dim, #class, and avg. #CLs denote the number of bags, number of instances, dimension of each instance, number of class labels, and the average size of candidate label set in each dataset, respectively. max #ins, min #ins, and avg. #ins denote the maximum, minimum, and average instance number across all bags, respectively. Moreover, CRC-MIPL is utilized, which is the first real-world MIPL dataset for colorectal cancer classification (CRC), comprising 7,000 hematoxylin and eosin staining images taken from colorectal cancer and normal tissues. The instances in CRC-MIPL are obtained by employing four image bag generators: Row, single blob with neighbors (SBN), k-means segmentation (KMeansSeg), and scale-invariant feature transform (SIFT), respectively. The candidate label sets of CRC-MIPL are provided by three crowdsourcing workers without expert pathologists.

AGOPMIPL is also compared with other baselines including five MIPL algorithms, two MIL algorithms, and two PLL algorithms. For the MIPL algorithms, MIPLGP (Tang et al., 2024b), DEMIPL (Tang et al., 2023), ELIMIPL (Tang et al., 2024a), ProMIPL (Yang et al., 2024), and FASTMIPL (Yang et al., 2025) are adopted. For the MIL algorithms, B-rFF (Pal et al., 2022) and Loss-ATTEN (Shi et al., 2020) are adopted, while for the PLL algorithms, PRODEN (Lv et al., 2020) and CAVL (Zhang et al., 2022) are utilized. Due to spatial limitations, only the MIPL experiment results are presented in the main body of the paper, and the results of MIL algorithms and PLL algorithms are detailed in the supplementary material. Parameters for all compared algorithms have been carefully tuned and set according to the recommendations in the literature, and the ratio of redundant noise labels is controlled by the parameter $r \in \{1, 2, 3\}$.

AGOPMIPL is implemented using PyTorch and trained on an NVIDIA A800 GPU. The optimization process employs stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 0.0001. For feature extraction, a two-layer convolutional neural network and a fully connected network are adopted for the MNIST-MIPL and FMNIST-MIPL datasets, while for the Birdsong-MIPL and SIVAL-MIPL datasets, a fully connected network is utilized to learn from the preprocessed features. For the CRC-MIPL dataset, a fully connected network following one of four image bag generators or ResNet is adopted as the feature extractor. The learning rate is selected from the predefined set $\{0.0005, 0.001, 0.002, 0.005\}$. The maximum number of training epochs is set to 200 for the MNIST-MIPL and FMNIST-

MIPL datasets, while those for the remaining datasets are set to 400. The data is divided into training and testing sets with a ratio of 7:3. The average and standard deviation of accuracy are recorded by conducting the experiments with random train/test splits ten times, and the highest accuracy is highlighted in bold.

Noise injection protocol. We make the candidate-noise level explicit. For each bag in the synthetic MIPL benchmarks, r false-positive labels are sampled *uniformly without replacement* from the set of non-true classes and added to the candidate set, while the unique ground-truth label is always retained. Hence the candidate set has size exactly $r + 1$, contains the true label, and contains no duplicates among false positives. On the 5-class MNIST-MIPL and FMNIST-MIPL datasets, this construction is well-defined only for $r \leq 3$ – the cases $r = 4$ and $r = 5$ would saturate the candidate set, and the corresponding stress test is therefore reported on the 25-class SIVAL-MIPL dataset (Section 5.4).

Key-instance evaluation metrics. We formalize how the ability to “find the key instance” is quantified in Figure 1 and in the experiments below. Given a trained bag-level model with attention scores α over the instances of a bag, let $\mathcal{K}_i \subseteq \{1, \dots, n_i\}$ denote the *key-instance set* of bag i , defined on the synthetic MNIST-style benchmarks as the set of instances that carry the true bag label.

- **Top-1 hit rate:** the indicator $\mathbb{I}(\arg \max_k \alpha_{i,k} \in \mathcal{K}_i)$ averaged over bags, i.e., the fraction of bags in which the highest-attention instance belongs to the key-instance set.
- **Average rank:** the mean rank, under descending attention, of the best-ranked element of $\mathcal{K}_i$, averaged over bags. Smaller is better.

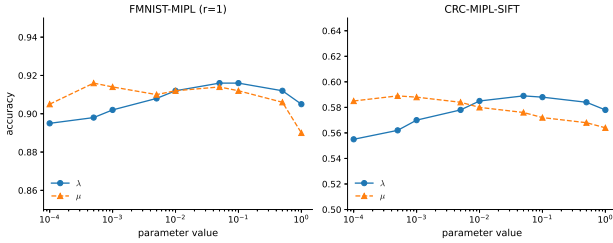
When a bag contains multiple instances that are positive for the true label, all of them are placed in $\mathcal{K}_i$ rather than only one designated representative; Top-1 hit rate then asks whether the highest-attention instance falls into this set, and Average Rank uses the best rank achieved by any element of $\mathcal{K}_i$.

5.1. Results

Table 2 presents a comprehensive comparison of AGOP-MIPL against five MIPL algorithms (FASTMIPL, ProMIPL, MIPLGP, DEMIPL, ELIMIPL) on benchmark datasets. The corresponding results of PLL and MIL algorithms are recorded in the supplementary material. From the results, it can be observed that AGOPMIPL achieves consistently higher mean accuracy across the four benchmark datasets

Dataset	#bag	#ins	max #ins	min #ins	avg. #ins	#dim	#class	avg. #CLs
MNIST-MIPL	500	20,664	48	35	41.33	784	5	2,3,4
FMNIST-MIPL	500	20,810	48	36	41.62	784	5	2,3,4
Birdsong-MIPL	1,300	48,425	76	25	37.25	38	13	2,3,4
SIVAL-MIPL	1,500	47,414	32	31	31.61	30	25	2,3,4
CRC-MIPL-Row (C-Row)	7,000	56,000	8	8	8	9	7	2.08
CRC-MIPL-SBN (C-SBN)	7,000	63,000	9	9	9	15	7	2.08
CRC-MIPL-KMeansSeg (C-KMeans)	7,000	30,178	6	3	4.31	6	7	2.08
CRC-MIPL-SIFT (C-SIFT)	7,000	175,000	25	25	25	128	7	2.08

Table 1. Experimental datasets with their characteristics

Figure 3. Performance of AGOPMIPL with varying μ and γ .

in most settings. Specifically, on the relatively easy MNIST-MIPL and FMNIST-MIPL tasks, AGOPMIPL achieves better or comparable performance, while for the more challenging Birdsong-MIPL, the proposed method is only slightly weaker than FastMIPL in the $r = 3$ case. On the SIVAL dataset, AGOPMIPL outperforms all five baseline methods. These results verify the effectiveness of the AGOPMIPL method.

Algorithm	r	MNIST	FMNIST	Birdsong	SIVAL
FastMIPL	1	.999 $\pm$.002	.911 $\pm$.022	.790 $\pm$.009	.708 $\pm$.030
	2	.998 $\pm$.004	.901 $\pm$.027	.770$\pm$.010	.630 $\pm$.026
	3	.975 $\pm$.074	.758$\pm$.071	.767$\pm$.022	.555 $\pm$.031
ProMIPL	1	.999 $\pm$.003	.921 $\pm$.024	.776 $\pm$.015	.682 $\pm$.032
	2	.999 $\pm$.003	.889 $\pm$.022	.719 $\pm$.018	.633 $\pm$.023
	3	.783 $\pm$.116	.659 $\pm$.041	.694 $\pm$.021	.539 $\pm$.024
EliMIPL	1	.991 $\pm$.005	.904 $\pm$.016	.770 $\pm$.019	.676 $\pm$.025
	2	.989 $\pm$.013	.843 $\pm$.026	.745 $\pm$.017	.615 $\pm$.023
	3	.749 $\pm$.148	.701 $\pm$.053	.717 $\pm$.019	.599 $\pm$.025
DeMIPL	1	.977 $\pm$.008	.883 $\pm$.019	.741 $\pm$.015	.631 $\pm$.042
	2	.944 $\pm$.027	.822 $\pm$.026	.702 $\pm$.026	.551 $\pm$.056
	3	.711 $\pm$.088	.656 $\pm$.027	.694 $\pm$.024	.502 $\pm$.017
MIPLGP	1	.951 $\pm$.019	.846 $\pm$.031	.714 $\pm$.026	.669 $\pm$.020
	2	.818 $\pm$.033	.792 $\pm$.027	.671 $\pm$.015	.614 $\pm$.023
	3	.623 $\pm$.062	.669 $\pm$.052	.626 $\pm$.015	.570 $\pm$.031
AGOPMIPL	1	.999 $\pm$.003	.922$\pm$.013	.806$\pm$.016	.759$\pm$.020
	2	1.00$\pm$.000	.904$\pm$.042	.769 $\pm$.043	.734$\pm$.017
	3	.986$\pm$.005	.752 $\pm$.013	.742 $\pm$.025	.705$\pm$.015

Table 2. Classification accuracy on MIPL benchmark datasets. Results are reported as best fold accuracy $\pm$ estimated std. The best result in each column is **bolded**.

Table 3 shows the experimental results on real-world

Algorithm	C-Row	C-SBN	C-KMeans	C-SIFT
ProMIPL	.435 $\pm$.009	.516 $\pm$.012	.565 $\pm$.013	.562 $\pm$.011
FastMIPL	.487 $\pm$.038	.573 $\pm$.031	.573 $\pm$.013	.526 $\pm$.029
EliMIPL	.434 $\pm$.008	.510 $\pm$.008	.545 $\pm$.013	.539 $\pm$.010
DeMIPL	.410 $\pm$.011	.484 $\pm$.013	.523 $\pm$.012	.531 $\pm$.013
MIPLGP	.435 $\pm$.006	.335 $\pm$.008	.331 $\pm$.014	—
AGOPMIPL	.534$\pm$.010	.610$\pm$.007	.722$\pm$.010	.589$\pm$.011

Table 3. Classification accuracy on CRC-MIPL datasets. The best result in each column is **bolded**.

Algorithm	Prediction Performance	AGOPMIPL Improvement
AGOPMIPL	.986$\pm$.012	—
FastMIPL	.975 $\pm$.074	win[$< 5e-2$]
ProMIPL	.783 $\pm$.116	win[$< 2e-4$]
EliMIPL	.749 $\pm$.148	win[$< 1e-3$]
DeMIPL	.711 $\pm$.088	win[$< 7e-9$]
MIPLGP	.623 $\pm$.062	win[$< 6e-6$]

Table 4. Summary of the Kendall tau correlation paired t-test across 5 labels on the MNIST-MIPL ($r = 3$).

datasets, where the proposed method achieves the best performance on all datasets. Compared with five existing MIPL methods, AGOPMIPL improves classification accuracy by 9.6% on C-Row, 6.4% on C-SBN, 4.8% on C-SIFT, and notably 25.9% on C-KMeans. We attribute this to two factors: (i) the prior assumptions used by ProMIPL and FastMIPL are harder to satisfy on the more complex real-world feature distributions, leading to their degradation; (ii) AGOP-based feature learning adapts the embedding metric directly to the data and, combined with feature prototypes, identifies key instances and disambiguates candidate label sets more effectively under such distributions.

Method	MNIST	FMNIST	C-KMeans
AGOPMIPL	1.00$\pm$.000	.752$\pm$.013	.722$\pm$.010
AGOPMIPL\A	.891 $\pm$.042	.621 $\pm$.031	.618 $\pm$.025
AGOPMIPL\P	.985 $\pm$.012	.698 $\pm$.018	.695 $\pm$.015

Table 5. Ablation study on the effectiveness of AGOP and progressive disambiguation ($r = 3$)

Moreover, the Kendall tau correlation paired t-test is per-

formed to verify the superiority of AGOPMIPL statistically; the results are shown in Table 4. AGOPMIPL outperforms all baseline methods on nearly all benchmark and real-world datasets, validating its effectiveness for MIPL tasks.

5.2. Ablation Study

To further verify the effectiveness of the proposed AGOP-based feature learning method and progressive label disambiguation strategy, ablation studies are performed on these two components. Experiments are conducted on the MNIST-MIPL dataset with $r = 3$, and the AGOP feature learning component and the progressive label disambiguation component are removed, respectively. The variants without these components are denoted as AGOPMIPL\A and AGOPMIPL\P, respectively. The results are shown in Table 5. The performance of AGOPMIPL\A drops sharply, validating the effectiveness of the AGOP component. Removing the progressive disambiguation strategy (AGOPMIPL\P) causes a smaller but still notable drop in accuracy.

5.3. Parameter Sensitivity Analysis

The two tunable loss weights μ and γ in Eqn. (7) are evaluated on MNIST ($r = 1$) and CRC-MIPL-SIFT (Figure 3); accuracy varies smoothly with both, indicating they are easy to tune.

5.4. Robustness under Heavy Candidate-Set Noise

The main results of Table 2 use $r \in \{1, 2, 3\}$, which is the regime in which the candidate set is non-degenerate on every benchmark. To probe robustness in a more demanding setting, we additionally evaluate $r \in \{3, 4, 5\}$ on the 25-class SIVAL-MIPL dataset, where the candidate set remains far from saturated even at $r = 5$. Following the noise-injection protocol above, r false-positive labels are sampled uniformly without replacement from the non-true classes for each bag. We compare AGOPMIPL with two competitive bag-level baselines (FastMIPL, ELIMIPL) under identical splits and report mean $\pm$ std over 3 folds.

Algorithm	$r = 3$	$r = 4$	$r = 5$
FastMIPL	.563 $\pm$.009	.531 $\pm$.009	.449 $\pm$.043
ELIMIPL	.453 $\pm$.016	.446 $\pm$.028	.409 $\pm$.023
AGOPMIPL	.693$\pm$.014	.654$\pm$.014	.544$\pm$.035

Table 6. Test accuracy on SIVAL-MIPL under heavy candidate-set noise. AGOPMIPL retains a substantial margin over both attention-based baselines as r grows.

As reported in Table 6, AGOPMIPL retains a sizable advantage over both baselines as r grows: even at $r = 5$, where the candidate set contains five false positives in addition to the

true label, AGOPMIPL outperforms FastMIPL by nearly 10 accuracy points. This is consistent with Proposition 4.2, which predicts that AGOP attenuates the contribution of bag-specific false-positive directions while preserving the cross-bag-coherent task-relevant subspace.

5.5. Additional Studies

AGOPMIPL is robust to its design choices. On the *prototype budget*, accuracy varies by less than 1.1 points across $n_p \in \{q/2, q, 2q\}$ on SIVAL-MIPL, so $n_p = q$ is a strong and stable default (Appendix C). On the *bag representation*, aggregating raw instance features $\mathbf{H}$ outperforms aggregating AGOP-transformed features $\mathbf{H}'$, indicating that AGOP is best used as attention guidance rather than a direct replacement for the aggregated content (Appendix D).

On the *computational side*, because $\mathbf{J}(\mathbf{z})$ is read analytically from a linear classifier head, the AGOP update costs only 0.17%–0.31% of one full training round across datasets ranging from 250 to 4900 bags, scales mildly with the bag-level feature dimension d' , and remains compatible with transformer-based instance encoders (Appendix E).

6. Conclusion

We propose AGOPMIPL, which uses AGOP to stretch discriminative feature dimensions and identify key instances under candidate-set noise, complemented by a progressive label disambiguation strategy. The key insight is that AGOP is computed by aggregating classifier sensitivities *across* all training bags: cross-bag-coherent true-label directions are amplified, while bag-specific candidate-noise directions are attenuated in expectation. This makes the learned feature metric inherently robust to noisy candidates without requiring an explicit denoising procedure. Extensive experiments show that AGOPMIPL outperforms existing MIPL methods on benchmarks and improves accuracy by 4.8%–25.9% on the real-world CRC-MIPL datasets. We believe the cross-sample-aggregation perspective on feature learning can serve as a useful design principle for other weakly supervised settings.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, with a particular focus on weakly supervised learning under candidate-set noise. The proposed AGOPMIPL method can help downstream applica-

tions – such as crowdsourced image annotation and medical histopathology classification – to extract reliable supervision from imperfect labels, thereby reducing the cost of obtaining accurately annotated data. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Training Procedure

The complete training procedure of AGOPMIPL is summarized in Algorithm 1.

Algorithm 1 Training Procedure of AGOPMIPL

Input: MIPL dataset $\mathcal{D} = \{(\mathbf{X}_i, \mathcal{S}_i)\}_{i \in [m]}$, hyperparameters μ, γ , EMA coefficient β , number of AGOP rounds R , epochs per round N

- 1: **Initialization:**
- 2: Collect bags according to candidate labels, form instance set for each class
- 3: Perform k -means on each class's instances to obtain n_p centroids $\mathbf{P}_c$ per class c
- 4: Initialize feature importance matrix $\mathbf{M}_0 = \mathbf{I}$
- 5: **for** $r = 1, \dots, R$ **do**
- 6: Reset optimizer with learning rate warmup
- 7: **for** $t = 1, \dots, N$ **do**
- 8: Compute confidence factor ρ_t via the warmup-decay schedule (Section 3.6)
- 9: **for** each mini-batch $(\mathbf{X}, \mathcal{S})$ in $\mathcal{D}$ **do**
- 10: Extract instance features $\mathbf{H} = \psi(\mathbf{X})$; form $\mathbf{E}$ by the learnable aggregation of $\{\psi(\mathbf{P}_c)\}_{c \in [q]}$
- 11: Transformation $\mathbf{H}' = \mathbf{H}\mathbf{M}_{r-1}^{1/2}$, $\mathbf{E}' = \mathbf{E}\mathbf{M}_{r-1}^{1/2}$
- 12: Compute dual-branch attention α (Eqn. 3–5)
- 13: Generate bag features $\mathbf{z} = \alpha\mathbf{H}$ (Eqn. 6)
- 14: Compute loss $\mathcal{L}$ (Eqn. 7) and update model
- 15: Update soft labels: $\tilde{\mathbf{y}}^{(t)} = \rho_t \mathbf{y} + (1 - \rho_t) \bar{\mathbf{p}}^{(t)}$
- 16: **end for**
- 17: **end for**
- 18: **if** $r < R$ **then**
- 19: Compute AGOP: $\mathbf{G}_r = \frac{1}{m} \sum_{i \in [m]} \mathbf{J}(\mathbf{z}_i)^\top \mathbf{J}(\mathbf{z}_i)$
- 20: EMA update: $\mathbf{M}_r = \beta \mathbf{M}_{r-1} + (1 - \beta) \mathbf{G}_r$
- 21: **end if**
- 22: **end for**

B. Proof of Proposition 4.2

Proof. (i) *Decomposition.* Expanding the AGOP definition with $\mathbf{J} = \mathbf{J}_{\text{true}} + \mathbf{J}_{\text{noise}}$ gives

$$\mathbf{G} = \mathbb{E}[\mathbf{J}^\top \mathbf{J}] = \mathbb{E}[\mathbf{J}_{\text{true}}^\top \mathbf{J}_{\text{true}}] + \mathbb{E}[\mathbf{J}_{\text{true}}^\top \mathbf{J}_{\text{noise}}] + \mathbb{E}[\mathbf{J}_{\text{noise}}^\top \mathbf{J}_{\text{true}}] + \mathbb{E}[\mathbf{J}_{\text{noise}}^\top \mathbf{J}_{\text{noise}}] = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top + 0 + 0 + \sigma^2 \mathbf{I},$$

where the mixed second moments are zero by Assumption 4.1, yielding Eqn. (9).

(ii) *Eigenstructure.* Let $\mathbf{V} \in \mathbb{R}^{d' \times (d' - s)}$ be an orthonormal basis for $\text{span}(\mathbf{U})^\perp$, so that $[\mathbf{U}, \mathbf{V}]$ is an orthonormal matrix in $\mathbb{R}^{d' \times d'}$ and $\mathbf{I} = \mathbf{U}\mathbf{U}^\top + \mathbf{V}\mathbf{V}^\top$. Substituting into Eqn. (9),

$$\mathbf{G} = \mathbf{U}(\mathbf{\Lambda} + \sigma^2 \mathbf{I}_s) \mathbf{U}^\top + \sigma^2 \mathbf{V}\mathbf{V}^\top,$$

which is a spectral decomposition of $\mathbf{G}$ with eigenpairs $\{(\lambda_i + \sigma^2, \mathbf{u}_i)\}_{i \in [s]}$ along $\mathbf{U}$ and the repeated eigenvalue σ^2 on $\text{span}(\mathbf{V}) = \text{span}(\mathbf{U})^\perp$. Since $\lambda_i + \sigma^2 > \sigma^2$ for all $i \in [s]$, the top- s eigenspace of $\mathbf{G}$ coincides with $\text{span}(\mathbf{U})$.

(iii) *Ratio bound.* Decompose $\mathbf{x} = \mathbf{x}_U + \mathbf{x}_V$ with $\mathbf{x}_U \in \text{span}(\mathbf{U})$ and $\mathbf{x}_V \in \text{span}(\mathbf{U})^\perp$. Using the spectral decomposition of $\mathbf{G}$ and the orthogonality $\mathbf{U}^\top \mathbf{x}_V = 0$, $\mathbf{V}^\top \mathbf{x}_U = 0$,

$$\begin{aligned} \|\mathbf{G}^{1/2} \mathbf{x}_U\|^2 &= \mathbf{x}_U^\top \mathbf{G} \mathbf{x}_U = \mathbf{x}_U^\top \mathbf{U}(\mathbf{\Lambda} + \sigma^2 \mathbf{I}_s) \mathbf{U}^\top \mathbf{x}_U \geq (\lambda_s + \sigma^2) \|\mathbf{x}_U\|^2, \\ \|\mathbf{G}^{1/2} \mathbf{x}_V\|^2 &= \mathbf{x}_V^\top \mathbf{G} \mathbf{x}_V = \sigma^2 \|\mathbf{x}_V\|^2, \end{aligned}$$

where the lower bound on $\|\mathbf{G}^{1/2} \mathbf{x}_U\|^2$ uses $\|\mathbf{U}^\top \mathbf{x}_U\| = \|\mathbf{x}_U\|$ and the fact that the smallest eigenvalue of $\mathbf{\Lambda} + \sigma^2 \mathbf{I}_s$ is $\lambda_s + \sigma^2$. Taking the ratio of square roots,

$$\frac{\|\mathbf{G}^{1/2} \mathbf{x}_U\|}{\|\mathbf{G}^{1/2} \mathbf{x}_V\|} \geq \frac{\sqrt{\lambda_s + \sigma^2}}{\sigma} \cdot \frac{\|\mathbf{x}_U\|}{\|\mathbf{x}_V\|} = \sqrt{1 + \frac{\lambda_s}{\sigma^2}} \cdot \frac{\|\mathbf{x}_U\|}{\|\mathbf{x}_V\|},$$

which is Eqn. (10). □

C. Prototype Budget Sensitivity

By default, each class is clustered into $n_p = q$ centroids, which are then aggregated into a single prototype per class; this centroid budget n_p is a practical convention rather than a theoretical requirement. To assess robustness to it, we vary n_p over a wide range relative to the number of classes q on SIVAL-MIPL ($q = 25$, $r = 3$ false-positive labels).

Prototype budget	Accuracy
$n_p = 12 \approx q/2$	.682±.014
$n_p = 25 = q$ (default)	.693±.014
$n_p = 50 = 2q$	.689±.017

Table 7. Sensitivity to the prototype budget on SIVAL-MIPL ($r = 3$ false-positive labels, 3-fold). Performance is stable across budgets, with $n_p = q$ remaining a strong and stable default.

Performance varies by less than 1.1 accuracy points across this $4\times$ range, indicating that the model is not narrowly dependent on the prototype budget. The default $n_p = q$ remains a strong and stable choice and is used throughout the rest of the experiments.

D. Bag Representation Choice

Eqn. (6) forms the bag representation by aggregating the *raw* instance features $\mathbf{H}$ with attention weights computed in the dual-branch attention. Two natural alternatives are to aggregate the AGOP-transformed features $\mathbf{H}'$ instead, or to concatenate the two. We test all three on SIVAL-MIPL ($r = 3$, 3-fold) and on the real-world CRC-MIPL-KMeansSeg dataset.

Bag representation	SIVAL ($r=3$)	CRC-KMeans
$\mathbf{z} = \alpha\mathbf{H}$ (default)	.693±.014	.7135±.0052
$\mathbf{z} = \alpha\mathbf{H}'$	.671±.016	.7135±.0052
$\mathbf{z} = [\alpha\mathbf{H}; \alpha\mathbf{H}']$	.680±.010	.7100±.0027

Table 8. Effect of the bag-representation choice. Replacing raw features with AGOP-transformed features in the aggregation hurts performance on SIVAL, suggesting that AGOP is better used as attention guidance than as a direct substitute for the bag feature; concatenation provides only a marginal change.

These results suggest that AGOP-transformed features are most useful as *attention guidance* – shaping which instances the bag-level pooling listens to – rather than as a direct replacement for the aggregated features themselves. Concatenation preserves some complementary information but does not exceed the simpler default in three-fold average.

E. Computational Overhead and Scalability

In the implementation of AGOPMIPL, the AGOP matrix $\mathbf{G}_r$ is computed on the bag-level representation $\mathbf{z} \in \mathbb{R}^{d'}$, and the practical overhead is therefore governed by d' , the number of training bags, and the AGOP update frequency, rather than by the number of raw instances or the input dimensionality d .

Overhead vs. dataset size. At fixed $d' = 128$, we profile the AGOP-update overhead on three representative datasets and contrast it with the full per-round training time. The AGOP update is performed analytically from the classifier head, so it does not require finite-difference passes or the storage of per-instance Jacobians (Section 4).

The AGOP overhead remains within 0.17%–0.31% of one training round across all three datasets, so the AGOP step is not a practical bottleneck even on the largest benchmark.

Overhead vs. feature dimension d' . We further sweep $d' \in \{64, 128, 256, 512\}$ on CRC-KMeansSeg (fold 1) to verify that the overhead grows mildly in d' .

Enhanced MIPL via Average Gradient Outer Product

Dataset	# bags	d'	Best Acc	Round-1 train (s)	Round-1 AGOP (s)
FMNIST-MIPL ($r=3$)	250	128	.712	133.53	0.42
SIVAL-MIPL ($r=3$)	750	128	.680	546.38	1.37
CRC-KMeansSeg	4900	128	.706	1970.02	3.40

Table 9. AGOP-update overhead vs. dataset size at fixed $d' = 128$. Across datasets ranging from 250 to 4900 training bags, the AGOP-specific overhead remains within 0.17%–0.31% of one full training round.

d'	Best Acc	Round-1 train (s)	Round-1 AGOP (s)	AGOP overhead	AGOP peak mem (MB)
64	.703	1972.23	3.42	0.17%	17.69
128	.706	1970.02	3.40	0.17%	19.05
256	.714	2021.05	3.61	0.18%	22.63
512	.705	2047.95	3.77	0.18%	30.28

Table 10. Sensitivity to the bag-level feature dimension d' on CRC-KMeansSeg. Increasing d' from 64 to 512 keeps performance essentially flat, while AGOP update time and peak memory grow only mildly.

Increasing d' from 64 to 512 keeps both accuracy and AGOP overhead essentially flat, with peak AGOP memory rising from 17.7 MB to 30.3 MB.

Feasibility on transformer-based encoders. The AGOP framework is not specific to CNN/MLP encoders. To verify that the pipeline remains trainable when the per-instance feature extractor is replaced with a transformer, we substitute the default encoder by a lightweight transformer (2 layers, 4 attention heads) and continue to apply AGOP at the bag-level d' -dimensional projection.

Encoder	Dataset	# bags	d'	Best Acc	Round-1 AGOP (s)
Default (CNN/MLP)	SIVAL-MIPL ($r=3$)	750	128	.680	1.36
Lightweight Transformer	SIVAL-MIPL ($r=3$)	750	128	.651	7.60

Table 11. AGOP at the bag-level projection layer remains compatible with a transformer-based instance encoder. The transformer variant is trainable and reaches 0.651 vs. 0.680 for the default encoder; the AGOP overhead remains a tiny fraction of total training time.

The transformer variant is trainable end-to-end and reaches 0.651 test accuracy versus 0.680 for the default encoder; the AGOP overhead remains $\sim 0.2\%$ of total training time. We do not claim that the transformer encoder is preferable here – the SIVAL features are low-dimensional tabular vectors for which CNNs/MLPs are well-suited – but the result demonstrates that the AGOP step itself is not architecture-specific.

Runtime and peak memory of AGOP itself. On CRC-MIPL-KMeansSeg (fold 1), one AGOP update takes about 19.0 s in round 1 and 3.7 s in round 2 with $\text{bag}_{\text{repr}} = \text{raw}$, and about 18.3 s and 3.8 s with $\text{bag}_{\text{repr}} = \text{rfm}$. Because the update involves only the bag-level head and is performed intermittently, the peak GPU memory dedicated to AGOP is small (Table 10). We therefore conclude that the AGOP step is computationally inexpensive in practice across the regimes considered.